RISC-V Event Trace Specification

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This document is in the Development state

Expect potential changes. This draft specification is likely to evolve before it is accepted as a standard. Implementations based on this draft may not conform to the future standard.

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Chapter 1. Introduction

Event Trace is an extension to the RISC-V Instruction Trace that provides hardware support for selectively tracing key executed events, called Event Types. It enables the capture of detailed execution information, which can be used for debugging, performance analysis, and system monitoring.

Event Trace extends the existing Instruction Trace while also leveraging its common packet types and trace sink options.

The Event Trace extension introduces new memory-mapped registers for setting up read-only contigurations, dynamical selection of trace events to trace, submodes of operation, hardware filters, and accessing its running status.

Chapter 2. Features

- Extension of ratified RISC-V Instruction Trace as an additional encoder mode and packet types
- Based on same core hardware interface, now defined as RHTI (RISC-V Hart Trace Interface) [1]
- A trace mode that turns off full instruction trace and turns on hardware filtering of instruction types and external interrupt events
- Event types that are independently traceable allowing control of the granularity of information captured
- Event types that include function calls and returns, exceptions, interrupts, context and privilege level changes
- Event types that include a timestamp based on a common fixed-frequency trace clock to provide precise timing information for an execution timeline and for performance analysis
- Event types that can selectively include profiling performance counter (PPC) values to provide core performance information such as IPC (instructions per cycle) or cache misses, summed for the duration of the event
- Run-time filtering of events based on type, privilege level, operating system context, and user-instrumented directives in source code to reduce the volume of trace data and to focus on relevant information
- External signals - edge-triggered signals that can insert a trace packet containing the closest retired PC when the event occurs, plus timestamp. This enables tracing of external events such as interrupts, I/O events, DMA start-end markers, uncore signals, or SoC-based events outside of program execution, and to time correlate them **with** program execution.
- Utilizes many of the capabilities and features of the existing RISC-V Instruction Trace thus ensuring compatibility to the existing Instruction Trace to reduce development costs and limit the hardware size of Event Trace extensions

2.1. Event Types

A list of the events types that can be traced with Event Trace is shown in the table below. Each type can be independently enabled or disabled for tracing.

Table 1. List of Event Trace Event Types

Name	Description
Function Calls, Returns	Calls are categorized as Inferrable and Uninferrable types, specifically: itype = 8 (indirect call), 9 (direct call), 12 (co-routine swap), 13 (return)
Exceptions	Occurrence (itype=1) records PC of instruction causing exception, cause register, + exit address (itype=3 (trap return))
Interrupts	Occurrence (itype=2) records PC of instruction that was interrupted, cause register, + exit address (itype=3 (trap return))
Trap Return	Occurrence (itype=3) records PC of return-from-trap instruction (MRET or SRET) of an exception or interrupt. This itype distinguishes the event from a normal function return.
Context + Priv Level	Records **s/hcontext** register when it is written to, plus **privilege level** and indication of s or h context

Name	Description
Debug trigger/watchpoints	Records PC of instruction that hit the trigger (or close to it depending on the type of trigger set), plus trigger ID and timestamp. Triggers and their comparison conditions are optional; they can include instruction retired at a specific address or range of addresses, data address read or write or access, data value, privilege mode, scontext, or other trigger conditions. Reference: [2 pp. Sdtrig].
User Definable Signal Events	Records latest retired PC when one or more user-definable signals are asserted. Section 3.5. signals are developer-defined, normally from the core, that are routed to the RHTI impdef[] (implementation-defined) bit array [1 pp. Optional sideband signals]. This feature provides a means to customize Event Trace to mark where and when the processor was executing when any of the signals are asserted. The trace packet includes all the signal values when any one goes true.
Periodic PC Sampling	Programmable timer that generates a trace packet with the most recent retired PC and timestamp when it expires. This allows for periodic PC sampling of the executing code to provide an execution profile over time.
External Signal (trigger) Events	Edge-triggered signals that can insert a trace packet containing the closest retired PC when the event occurs, plus timestamp. This enables tracing of external events such as interrupts, I/O events, DMA start-end markers, uncore signals, or user-selectable SoC-based events outside of core execution and to **time correlate** them with program execution. External Trace Triggers are defined in the Trace Control Interface spec [3 pp. External Trace Triggers].
Performance Counter Selector Events	RISC-V HPM (Hardware Performance Monitor) [4] includes selector registers (mhpmeventX) which are user-programmed to select one (or several) core event types to count. This Event Trace feature extends this performance counter feature to record the closest retired PC on the rising edge of mhpmeventX output and save it in the trace buffer thus showing **where** in the code the core event occurred.

2.2. Timestamp

The **Event Trace Timestamp** is the same timestamp timer used by Instruction Trace. It is a continuous running count-up counter (not a delta timer). As the RISC-V-Trace-Control-Interface specification describes in detail, there are optional sources for the timestamp: 1) External, 2) Internal System, 3) Internal Core, and 4) Shared.

If the (optional) Timestamp Unit is included in the Instruction Trace, then:

- **Timestamp enable**: must be enabled by setting **trTsControl.trTsActive**, **trTsControl.trTsEnable**, and started by setting **trTsControl.trTsCount**.
- **Timestamp off**: turned off, i.e. not included in trace packets (for Instruction AND Event Trace) by clearing **trTsControl.trTsEnable**. This can save some trace bandwidth and storage.
- **Timestamp reset**: timestamp register can be reset by setting, then clearing **trTsControl.trTsReset**. This is recommended before starting a trace session since lower-values timestamps can benefit trace bandwidth and storage, because of leading-zero suppression.

It is highly recommended that the timestamp clock run at or a close ratio to cpu clock in order to accurately time short functions and point-to-point events such as hardware triggers.

2.3. Profiling Performance Counters (PPCs)

Event Trace can include the option of Profiling Performance Counters (PPCs) which are delta counters that are recorded at the time of an event, simultaneously reset to zero, then begin counting again. This is a requirement to keep the counter values smaller to reduce trace bandwidth and storage. Post-processing of

the trace data can reconstruct event counts across multiple event sequences to provide accumulated totals. An example would be inclusive function call-return counts, where snippets of code execute in a function then more counts for functions that it calls. "Self" or "exclusive" counts, i.e. only counts attributed to the code of the function itself would only accumulate counts from calls to other functions and from returns to calls of other functions.

2.4. Event Trace Block Diagram

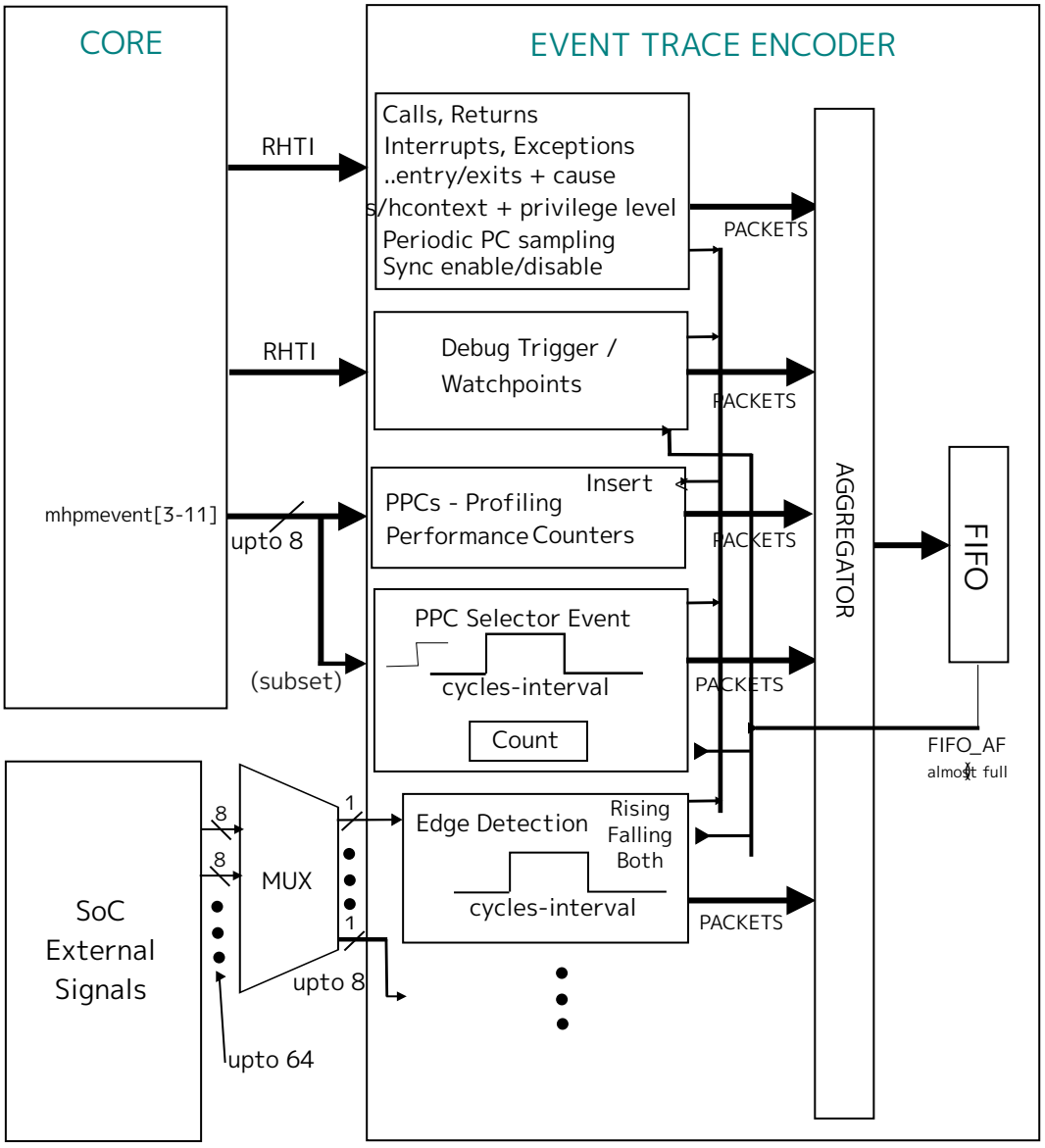


Figure 1. Event Trace Block Diagram

2.5. Features and Benefits (non-normative)

All Event Trace features are optional, however, the minimal set of event types recommended are: call/returns, interrupts, exceptions and context ID + privilege level.

Table 2. Event Trace Features and Benefits

Feature	Benefits
Function Calls, Returns	Trace history of function calls/returns assists in debugging and making performance measurements, all with zero execution overhead. Trace can be compressed and visualized as a flame graph. Timestamping provides complete timing analysis of function execution including min, max, and average durations and the number of times a function has been called. The addition of Profiling Performance Counters (PPCs) provides detailed microarchitectural information about the dynamics of processor efficiency and throughput such as IPC, cache misses, branch mispredicts, etc., with measurement granularity down to the function level.
Exceptions	Tracing occurrences of exceptions and their causes can be beneficial for locating illegal instructions, instruction or data misalignments (that slow down execution speed), and page faults. These types of code conditions may occur infrequently such that the may be difficult or impossible to trace with generic instruction trace. Exceptions occur whenever the processor undergoes privilege mode changes; tracing these can show when and for how long these privilege levels take to execute. Also includes is the tracing of Linux system calls.
Interrupts	Tracing occurrences of interrupts and their causes can be beneficial for observing correct interrupt priorities, nested interrupts, and the duration of interrupt handlers - especially critical in embedded systems.
PPCs - Profiling Performance Counters	PPCs provide detailed microarchitectural information about the dynamics of processor efficiency and throughput such as IPC, cache misses, branch mispredicts, stalls, etc. They can be used with Watchpoints to measure precise execution from point A to B of a code segments.
Context + Privilege Level	A high level view of execution can be captured by tracing context and privilege level changes. These show timing of each process and time executing the OS kernel as well as low-level hardware accesses in M-mode. A pie chart can show time in each privilege level. Waveforms can show the sequence of program and OS executions.
HW Filters for Privilege, Context ID, and Exception and Interrupt Cause	Filtering benefits the utilization of trace memory by recording only events that are relevant to the debugging or performance analysis task. For example, if the developer is only interested in user-level code, then filtering out events that occur in M-mode and S-mode can save trace memory and make post-processing more efficient. If the developer is only interested in a specific process, then filtering by context ID excludes all other processes, including the OS itself. If the user is only interested in specific exception types or interrupt causes, then the resulting display will show only relevant data.

Feature	Benefits
Debug trigger/watchpoints	Tracing hardware trigger/watchpoints [2 pp. Sdtrig] are useful for programming markers without having to instrument and recompile code. They provide traces of specific locations in one's code for timing executions, counting occurrences, and especially for precisely measuring algorithic loop times, variations, and observing data-dependent behavior, such as cache misses or a function that executes in a varying amount of time based on the input data. For device driver debugging, watchpoints can be set on memory-mapped I/O addresses to trace reads or writes for debugging and performance throughput of I/O. These types of measurements are difficult or impossible to trace with generic instruction trace because of the infrequency of the events and the fact that they may be accessed by a number of different locations in the code.
Periodic PC Sampling	Periodic PC sampling provides statistical profiling of code execution over time, with no performance penalty that software-based sampling incurs. This mode can very quickly identify "hot spots" in the code as candidates for optimization. With the addition of profiling performance counts, this trace mode provides details of program conditions accumulated at each of the sample addresses.
External Signal (Trigger) Events	Applicable for embedded systems, tracing external events such as the rising edge of an interrupt signal, the closest concurrent retired instruction along with tracing the cpu processing of the interrupt - this combination provides a means of precisely measuring interrupt latencies. Capturing many of these will characterize the timing and variability of interrupt processing and could expose maximum values that violate real-time requirements. External signal tracing is primarily for a few (up to 8) signals that don't transition rapidly (i.e. not for tracing buses). The user has the choice to record only rising, only falling, or both edges of a signal, the latter for measuring pulse widths. Each edge generates a trace packet with timestamp which can be shown as a "logic analyzer" waveform view. This can provide visibility into signals and timing that may otherwise be impossible to observe.
Performance Counter Selector Events	Normally, performance counter selector registers (mhpmeventX) are used to select one or more core events to count. With this Event Trace feature, the closest retired PC is recorded in the trace buffer on the occurrence of the mhpmeventX output signal. This shows **where** in the code the core event occurred; post-processing these addresses into module/function/line numbers provides direct user feedback as to the code locations where performance bottlenecks are occurring. For cpu events that can occur in high-frequency bursts such as cache misses, the trace packet captures a count of the number of occurrences of the event over a user-selectable interval of time.

Chapter 3. Event Types

3.1. Non-function Events

3.1.1. Event Trace Message Fields

F-ADDR *means full or uncompressed or synchronized address.* **C-ADDR** *means compressed address which the trace decoder converts into a full address from a previous packet, starting from a sync packet that includes a full address. A C-ADDR, for N-Trace [5], is called a U-ADDR (U=unique) which is generated with "previous PC XOR current PC" to compress. For E-Trace [6], compressed addresses (or delta address mode) are referred to as a differential encoding which is hardware logic that does "previous address - current address".*

Table 3. Event Trace Message Fields for non-function events

Event	EVTPC	EVTDATA	ITYPE	Description
Sync Enable	PC	SYNC	X	SYNC field: EVTPC is always F-ADDR Causes for Sync Enable: * Exit from Reset * Exit from debug. PC=first instruction to execute after debug mode. * Event trace enable. PC=instruction responsible for trace-on.
Sync Disable	PC	EVCODE	X	EVCODE field: EVTPC is always F-ADDR Causes for Sync Disable: * Enter-debug. PC=address loaded into depc (has not executed). * Event trace disabled. PC=instruction responsible for trace-off. * Enter-reset. PC=last instruction to execute before reset.
Exception	PC	cause	1	PC (C-Addr) is what is loaded into xepc (the address of the instruction that caused the exception), **cause** is what is loaded into the Exception Code field of xcause. cause signals are defined in Table 1 of RHTI spec [1].
Interrupt	PC	cause	2	PC (C-Addr) is loaded into xepc, **cause** is what is loaded into the Exception Code field of xcause.
Return from trap	PC	PCdest	3	Return from exception or interrupt (MRET or SRET) (to distinguish from normal function return) PC and PCdest (both are C-ADDR) are both generated when they are not filtered by a privilege mode filter.
Return from trap	------	PCdest	3	(Variant of above) Return from exception or interrupt (MRET or SRET). Generated when the source PC is filtered by privilege mode (but PCdest which is a C-ADDR is not).

3.2. Function Events

The following table lists the packet types to support function Calls and Returns, with different types and numbers of parameters.

Table 4. Event Trace Message Types for Calls and Returns

Name	Source	Destination	ITYPE	Description
CALL_SYNC	PC (F-ADDR)	dest (C-ADDR)	8 or 9	Traces both src and dest addresses. The second address can be compressed. Full (absolute) timestamp (if TS enabled).
CALL_FULL	PC (C-ADDR)	dest (C-ADDR)	8 or 9	Traces both src and dest compressed (unique) addrs; relative timestamp (if TS enabled).
CALL_DEST	------	dest (C-ADDR)	8	itype=8 means indirect (uninferrable) call or co-routine swap. Source function can be derived from trace history however src address (i.e. line # of caller) is unknown.
CALL_PC	PC (C-ADDR)	------	9	inferrable call so dest obtained from opcode if binary available
RET_FULL	PC (C-ADDR)	dest (C-ADDR)	13	Traces both src and dest (return to) addresses
RET_DEST	------	dest (C-ADDR)	13	return-to address; caller (function) determined from symbolic address range of destination address and/or from history.
RET	------	------	13	Function returned to is determined from history. If caller was CALL_PC (inferrable caller), decoder can determine return destination address from 2-byte or 4-byte opcode.

If Calls/Returns are supported, Event Trace requires implementation of CALL_SYNC, CALL_FULL, and RET_FULL. CALL_DEST and CALL_PC are useful for reducing bandwidth. RET_DEST and RET are also useful for reducing bandwidth.

3.3. Context + Privilege Level Events

Instruction Trace already has the capability to trace changes in context ID and privilege level. Event Trace will utilize the same packet type to trace these events - for N-Trace [5] and E-Trace [6].

ORing of controls for enabling Context and Privilege level

- Set **trTeControl.trTeContext** of Instruction Trace to enable tracing of Context changes, or
- Set Event Trace Context enable control bit **trTeControl.trTeEvtSContextPrivEn** (when Supervisor Context CSR written to) and **trTeControl.trTeEvtHContextPrivEn** (when Hypervisor context CSR written).

3.4. Debug Trigger/Watchpoint Events

A Debug Module can (and most often does) include a Trigger Module [2 pp. Sdtrig] - hardware which matches on core conditions such as instruction address, data address, data value, privilege mode, context ID, etc. The Trigger Module can be programmed to generate an action called **Trace-notify** [2 pp. Tb. 12] using the value 4 for the 4-bit action field which is then routed to the RHTI [1] defined **trigger [2]** signal. Refer to Section 7.1 section for details.

For Event Trace, the **Trace-notify** action, when programmed by the user and enabled in Event Trace, results in a packet packet output that includes the PC of the instruction that hit the trigger/watchpoint and

a timestamp, plus PPCs if enabled. Thus Event Trace will record these Debug Triggers, and depending on the specific implementation of hardware debug triggers, can be amarker for instruction address/addresses executed or when certain data addresses have been accessed.

The RHTI Extensions section Section 7.1 defines the **triggerID** signals for identifying which programmed trigger/watchpoint condition was met. The **triggerID** value will be included in the trace packet to identify which trigger/watchpoint caused the trace event.

If the triggerID is not provided, trace decoder software may be able to identify and fill in the triggerID in the trace output. This first requires saving the trigger comparator setup for each trigger - primarily the execution or data address comparison value. The trace decoder, with this information, can then match the recorded PC in the trace packet to the list of programmed trigger values and fill in the triggerID.

3.5. User Definable Signal Events

Signals are developer-defined, normally from the core, that are routed to the RHTI impdef[] (implementation-defined) bit array [1 pp. Optional sideband signals]. This feature provides a means for a developer to customize Event Trace to mark when and where the processor was executing when any of the signals are asserted. The trace packet includes all the signal values when any one goes true.

3.6. Periodic PC Sampling Events

For the **Periodic PC sampling** event type, a programmable counter runs until it expires which generates a trace packet with the most recent retired PC and timestamp then is preloaded with the count and counts down again. This allows for periodic PC sampling of the executing code to provide an execution profile over time. Note that with the inclusion of profiling performance counters (PPCs), this trace mode provides details of program conditions accumulated at the sample addresses.

While Instruction Trace provides a counter for periodic sampling, its range does not the best match for Event Trace. Therefore Event Trace will provide its own counter for periodic PC sampling. It uses the same powers-of-2 (exponential) sample count, with a count of **2^(<value>+6)**. With a 4-bit <value> selection field (0 to 15), the range extends from 2^6 to 2^21, or 64 counts to 2 million counts. Assuming counts = cpu cycles, the range is 64 cycles (64ns at 1GHz) to 2 million cycles (2ms at 1GHz) which is practical for sampling code execution.

Randomizing PC sampling. To avoid aliasing of the periodic PC sampling of a repeating code execution pattern, hardware can randomize the sampling interval by including an additional random count after the powers-of-2 count has counted down. One method is to use a linear feedback shift register (LFSR) to generate a pseudo-random number which is used to pre-load a final count-down counter.

The number of randomizing bits, i.e. width of the LFSR is developer-selected. (Six is practical). Note that for shorter intervals the randomizing has a larger percentage effect on the interval than for longer interval durations.

3.7. External Signal Events

Features:

- 8 external input signals, also called triggers or channels
- recognize the rising edge on an input signal and generate a packet
- (optional) user-programmed to recognize a falling edge
- (optional) user-programmed to recognize either a rising or falling edge on the input signal, useful for timing the duration of a signal pulse.
- includes a **cycles-interval** to compact multiple edges into one trace packet. This is to prevent overloading the trace system with too many events in a short period of time when the signal is changing frequently. The logic, after desired edge is detected, begins the packet creation by latching the edge value and timestamp. If, during the cycles-interval, one or more additional signal edges occur, the packet records the signal as having multiple edges. When the cycles-interval expires, the packet is formatted and output to the trace sink.

(Non-normative) The post-processing visualization (e.g. logic analyzer-like timing waveforms) shows the multi-edge packet as a "fuzzy" unit meaning it represents > 1 edge. The cycles-interval count would indicate the relative width of this "fuzzy" edge. The recorded timestamp, however, does show the time location of the starting edge of the signal.

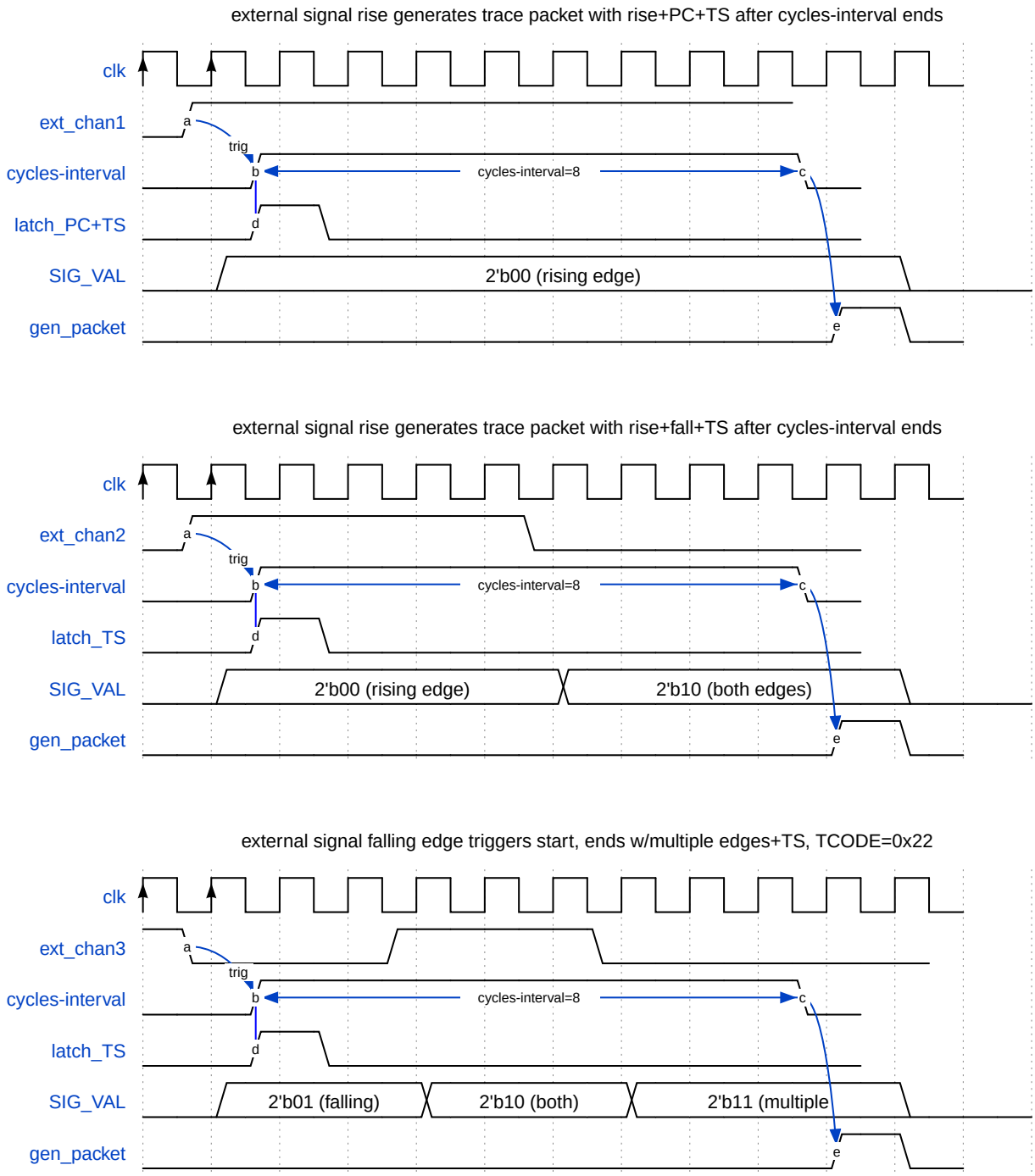
- **cycles-interval** uses a 4-bit powers-of-2 selector to count-down a counter, with a range from 2 to 2^15 (32,768) cycles duration.
- implement a trace packet that records 1) rising edge or 2) falling edge, or 3) either edge
  - which value recorded is dependent on user request for which edge/edges) to trace
  - or 4) > 2 edges within the interval window
- provide two trace packet types:
  - rising, falling, both, or multiple edges occurred + channel ID + timestamp at the time the initial edge was observed. (NOT recording PC when it is not needed, saving bandwidth and storage).
  - rising, falling, both, or multiple edges occurred, channel ID, timestamp, and **PC** of the closest retired instruction at the time the edge was observed
- enable/disable the recording of each input signal, independently
- multiplexer selector control to support from 1 of 64 sets of 8 inputs into the hardware trace logic

The implementer must be aware that each channel is independent and has its own edge detection, cycle interval counter, latching of edge state, timestamp and PC (if enabled). Events like a rising edge could occur simultaneously on multiple channels, including all channels. The trace encoder must serialize these events into a packet stream and into trace storage.

3.7.1. Waveform examples of tracing an external signal event

This section includes three examples of external signal edge capture within the cycles-interval time window - rising edge, falling then rising edge, and three (multiple) edges.

- each example uses a cycles-interval of 8 cycles (2^3)
- at the detection of the first edge, which takes two sample clocks, "latch_PC+TS" saves the closest retired instruction PC and timestamp. "latch_TS" option only records the timestamp.
- SIG_VAL is the value of the signal at the time of the first edge, and if additional edges occur within the cycles-interval, it records either "both edges" or "multiple edges" have occurred. *gen_packet is the time in the recording when the trace packet is generated, which is at the end of the cycles-interval. The packet includes the channel ID, timestamp, and PC (if enabled) and the signal value (rising, falling, both, or multiple edges).



3.8. Performance Counter Selector-Based Events

RISC-V programmable performance counters have selector registers (programmed with the mhpmeventX CSRs) which are user-programmed to select one (or several) event types to count. (The 'X' of mhpmeventX are numbered 3-31). The selector outputs a "1" level when a certain countable event occurs such as a cache miss, branch mispredict, or stall.

Single event vs. multi-cycle duration. Some counter event types are singular events - they are true for one cpu cycle indicating an occurrence such as an intruction retired. Some event types are true for multiple cycles indicating a condition that stays true for a duration, where a performance counter accumulates that number of cycles while the condition is true.

Level-to-Edge Conversion. This document assumed selector outputs are level sensitive. The Event Trace hardware requires edge sensitivity, thus it needs to convert the level to an edge to trigger an event to be recorded by Event Trace. The occurrence of the event can remain at a high level but one one instance of the event is recorded. (Another way of describing this is to convert a multi-cycle level into a single-cycle pulse).

Stretching logic high levels. Event Trace hardware must accommodate signals that can change level on every CPU cycle. If the clock used for the Trace Encoder runs at a divided ratio to the CPU clock, then the Event Trace hardware may require stretch logic to convert single-cpu-cycle levels into a level recognizable by the edge detector.

Single value output. Another programmable performance counter attribute is that some events report greater than 1 value for a single cycle. This can occur, for example, when counting retired instructions or even a class of instructions like loads that may retire more than one per cycle. A simplification for Event Trace is to treat all selector outputs as a single value of 0 or 1 (converting > 1 to 1 which can be done with an OR gate). An asserted selector output can be thought of as a code marker.

Tracing perf counter events. These (mhpmeventX) selector outputs are routed to the Event Trace hardware to generate a trace packet (when enabled). The trace packet latches the most recent retired instruction PC and timestamp. This enables tracing of where in the executing code a performance event occurs such as a cache miss, branch mispredict, or any other countable event that can be selected by a performance counter selector.

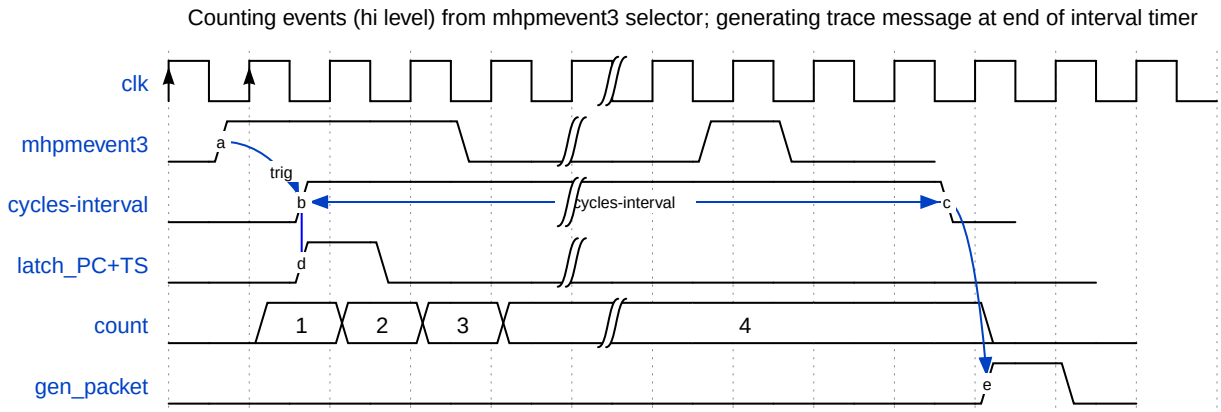
Enabling selector recording. An Event Trace control register Table 13 includes an enable bit for each performance counter selector included in Event Trace.

There are a class of programmable performance counter events that can occur with high frequency. To avoid generating a trace event for every one which could overload the trace system, the selector event hardware includes a **cycles-interval** timer [select-cycles-interval] which programs an interval in which the hardware counts occurrences of the event. At the end of the interval, Event Trace constructs a packet that includes the selector ID, the latched timestamp when the first event occurred, and the count. The duration of the cycles-interval timer is user selectable, with a minimum and maximum value.

The counter is defined to be 24 bits (increments-of-6 work best for N-Trace [5]), and is a saturating counter so that it does not roll over and cause loss of information.

Control registers for perf counter selectors. Event trace control register includes a Read-Only field that specifies how many performance counter selectors are included in the Event Trace implementation [SelectorsImplemented]. A practical number is, however, between 2 and 4.

3.8.1. Waveform example of tracing a performance counter selector event



- **mhpmevent3** is the signal being traced, from the output of the performance counter selector. It goes high when the selected event occurs; the rising edge is the start of the event recording, and the logic then counts its high state while cycles-interval is true.
- The **cycles-interval** timer is triggered on the rising edge (zero to one sampling)of the mhpmevent3. When the timer ends, this triggers the gen_packet to generate the trace packet.
- **latch_PC+TS** is the signal to latch the most recent retired PC and timestamp on the rising edge of mhpmevent3. This is the value that is recorded in the trace packet when the timer expires and the packet is generated.
- **count** is the count value while mhpmevent3 level is high and cycles-interval-timer is true.
- **gen_packet** is the signal to generate the trace packet at the end of the cycles-interval timer. It also resets the count to zero and prepares for the next event.

3.9. Limiting Event Trace Bandwidth based on Trace Encoder Feedback.

Most Trace Encoders will include a FIFO for buffering trace events and to smooth out bursts of trace recordings that can occur at higher rates than the back-end rate of streaming trace packets to memory or PIB (probe interface block). Most FIFOs have a buffer depth signal indicating when it is close to full. This indicator, here labeled **FIFO_AF** (almost full), can act as a back-pressure signal to the Event Trace hardware to temporarily stop the generation of trace packets.

This is a front-end throttle to prevent the Trace Encoder FIFO from filling up and using the Encoder back-end to stop tracing (which generates an error packet when tracing is resumed). The lockout affects all external inputs to the encoder (when enabled) including Debug trigger/watchpoints Section 3.4, User Definable Signals Section 3.5, External Signals (triggers) Section 3.7, and Performance Counter selectors Section 3.8. The control register to enable or disable this feedback mechanism is [EventLockout].

Chapter 4. Profiling Performance Counters (PPCs)

- PPCs are defined in Event Trace as delta counters, counters that are recorded at the time of an event, simultaneously reset to zero, then begin counting again. This is a requirement to keep the counter values smaller to reduce trace bandwidth and storage. Post-processing of the trace data can reconstruct event counts across multiple event sequences to provide accumulated totals. An example would be inclusive function call-return counts, where snippets of code execute in a function then more counts for functions that it calls. "Self" or "exclusive" counts, i.e. only counts attributed to the code of the function itself would only accumulate counts from calls to other functions and from returns to calls of other functions.

Two mechanisms for implementing delta counters are:

- Delta counters based on hpmcounters, or
- Delta counters based on independent (Event Trace) counters, i.e. not reliant on hpmcounters

Implementers have the option to implement one or the other these mechanisms.

4.1. Alternative 1: Delta counters based on hpmcounters

The first mechanism may appear to save hardware costs by using existing hpmcounters, however the disadvantages can be:

1) delta counts are generated when a new event occurs, which terminates the previous event and the delta count is assigned to that terminated event. Generating delta counts requires additional latching of the running performance counter at the end of the terminating event and the beginning of the next event so it can be subtracted from the counter at the time that event terminates, to produce the delta count.

2) these counters become a shared resource between traditional software-controlled performance counters and their use with Event Trace. In other words, if Event Trace is being used with these counters, they are not available to be used as traditional performance counters for software to read and write.

3) hpmcounters are CSRs and located in the core. To implement delta counts requires either routing the outputs of each of these counters to the Trace Encoder to latch and subtract the start and end values or, adding this logic into the core and routing the delta count to the Trace Encoder. Both options require additional wires and routing complexity; the latter option of generating the delta counters in the core then requires only 24 wires per counter to be rounted to the Trace Encoder.

No additional control registers are required for this configuration since the hpmcounters are being used to generate the delta counts.

4.2. Alternative 2a: Delta counters based on independent Event Trace counters (not reliant on hpmcounters), not reliant on mhpmeventX selectors

For Event Trace, independent profiling performance counters can be implemented as delta counters, with the output latched then reset to zero at the start of the next event. These delta counters saturate at their maximum all-ones value to avoid roll-over.

Alternative 2a, independent PPCs, also implement their own event selectors (see Section 3.8) and thus make them completely indpendent of the HPM unit.

The previous section covered the Event trace feature "Event Trace Performance Counter Selector-Based Events" which generates a trace packet when an enabled performance counter selector output is a "1" level. The PPCs are a way of counting selector events and capturing the summed counts at the termination of an event, thus compacting these occurrences into a count. If this feature is implemented then the same selector outputs that generate Event Trace Performance Counter Selector-Based Events can also be routed to the PPCs to count the occurrences of these events between Event Trace events.

4.3. Alternative 2b: Delta counters based on independent (Event Trace) counters (not reliant on hpmcounters), however reliant on mhpmeventX selectors

Alternative 2b, independent PPCs, relies on HPM event selectors (mhpmeventXs) routing the outputs of the mhpmeventX selectors in the HPM module to the Event Trace unit.

This option saves hardware costs but requires the sharing of performance counter resources of the HPM unit. Note that this sharing still allows for performance counters to be used at the same time as Event Trace uses them. It just means that the settings for these mhpmeventX selectors have to be the same for both uses.

4.3.1. Recommended features of Profiling Performance Counters (PPCs) for Event Trace (non-normative)

- 24-bit delta counters, max of 8
- the number of counters implemented is in pairs of 2, thus the number of PPCs is 2, 4, 6, or 8
- counters are saturating, i.e. stop counting at all-ones max value until the next event occurs and resets each counter to zero. This provides a max count of 2^24, i.e. 16 million
- map each PPC to a MMIO address to accommodate testing and DV of the hardware.
- as a secondary use, provide a control register field to latch PPCs thus allowing JTAG control and readout of PPCs periodically. Since they are memory-mapped, this performance counter sampling can be done with no cpu intervention which is not possible with CSR-based performance counters.

4.4. Fixed Counter - Instructions Retired

A very common and useful performance measurement is IPC - instructions per cycle. This ratio indicates the instruction throughput of the core and identifies the actual code performance relative to the ideal. For example, if a core is a 2-issue/2-instruction retire per cycle core then its ideal and maximum performance would be IPC=2. Numbers less than 2 indicate how far from the ideal it is performing. Measuring IPC per function with Event Trace provides a reasonably-fine granular view of performance, and sorting function results by IPC will identify which functions are furthest from the ideal and thus have the most to gain by improving their performance.

The current RHTI specification [1] does not include a signal set for reporting the number of instructions retired in each cycle. The proposal for adding this signal set is here - Instruction Retired Count.

An implementer can include the **Fixed counter - Instructions Retired** feature without the RHTI spec by accessing and routing these signals from the HPM unit to the Trace Encoder.

Counting instructions retired is a RISC-V requirement, which is the **minstret** *CSR*

> *(instructions retired is hpmcounter1), a 64b performance counter (section 3.1.10 Hardware Performance Monitor, RISC-V Privileged Architecture Manual). In other words, it is already implemented in the core. The signal set simply needs to be routed to the Event Trace Encoder.*

If the developer is using Alternative 2 (independent counters), one choice is to dedicate the first 24-bit counter for counting retired instructions between events, with the same delta saturating counter mechanism as other PPCs. This provides a max count of 16 million instructions between events. This counter must be able to count > 1 instruction per cycle if the core retires more than 1 instruction per cycle. One disadvantage of this is that the **minstret** counter does not support count filtering by privilege level whereas a programmable counter can, if a selector (mhpmeventX) has a value that includes counting instructions retired.

> **Counting cycles.** *For IPC ratio calculations, the number of cycles between events is also required. The timestamp may or* **may not** *be counting cpu cycles. If it does not, the Event Trace Decoder must be informed what the fixed ratio of timestamp ticks to CPU cycles is so it can convert timestamp counts into cycle counts. One hardware method for recording the timestamp-to-cycle ratio is to implement the Time Sync Packet.*

Chapter 5. Event Trace-specific Hardware Filtering

5.1. Leaf Function Filtering

Leaf function filtering is the hardware filtering of call-return pairs which meet a time duration "less-than N cycles" threshold in order to reduce trace bandwidth and save storage for short functions that are less interesting to trace.

More documentation to follow.

Chapter 6. Control Register Map

6.1. Control Register Map, Summary

Event Trace as a trace mode (separate from Instruction Trace) is set by software writing trTeControl.trTeInstMode=4 which is defined in the N-Trace spec [5 pp. Protocol defined trace mode].

Event Trace register starting point is an offset from the Trace Encoder base address. The Event Trace allocates and uses addresses **0x100-0x1FF**. Thus trTeEvtBase = trBaseEncoder+0x100. In the table below, the Offset is from the trTeEvtBase.

Table 5. Event Trace Control Register Map (Summary)

Name	Offset	Description
trTeEvtControl	0x0	* EVT tracing status * EVT trigger enable * call/return modes * Event per-type enables * periodic sampling duration * enable event lockout
trTeEvtFeatures	0x04	Defines which Event Trace features are implemented, and sizes.
trTeEvtMissed	0x08	Counter which is incremented whenever an event is discarded because the Encoder cannot accept the event.
trTeLeafFilterControl	0x0C	Enables leaf function filtering and sets the threshold for filtering out call-return pairs that are less than N cycles in duration. The threshold is set in powers-of-2, i.e. 2^N cycles, where N=0 to 15. A value of 0 disables leaf function filtering.
trTeEvtPPCEnables	0x10	Defines which Event Types output Profiling Performance Counter (PPC) values. Sets the number of PPCs output with each Event Type enabled.
trTeEvtPPCControl	0x14	Control register that, when written to, causes all PPCs to be latched. Software can then read out all the PPC values. Only active when Event Trace is disabled.
Reserved	0x18	
trTeEvtPPCCounter*i*	0x20, 0x28, 0x30, 0x38, x040, x048, 0x50, 0x58	24b PPC Counter access registers (if developer chooses Section 4.2 or Section 4.3 - independent PPCs)
trTeEvtPPCEvent*i*	0x60, 0x68, 0x70, 0x78, x080, x088, 0x90, 0x98	64b PPC Event Selector Registers (if developer chooses independent selectors vs using mhpmeventX selectors). For XLEN = 32, the upper 32b of each register are accessible with the address of trTeEvtPPCEvent*i* + 4.
trTeEvtExtSigControl	0xA0	Control register that enables up to 8 external signals for tracing signal transitions (edges)
trTeEvtPerfSelectControl	0xA4	Control register to enable tracing of performance counter selector events and setting an interval for counting occurrences while true.

6.2. Event Trace Control Register Map, Details

6.2.1. trTeEvtControl: Event Trace Control Register (trBaseEncoder+0x100)

Table 6. Event Trace Control Register fields for **trTeEvtControl** *(32b)*

Bits	Field	Description	RW	Reset
0	trTeEvtImplemented	Read as 1 if Event Trace is implemented	RO	1
1	trTeEvtTracing	(Read) EVT **status**: 0: not tracing; 1:event trace is running. HW Triggers (in the Trigger Module)can enable or disable EVT. (Write) EVT **control**: When **trTeEnableTracing** = 1, writing 0 will disable trace that then can be enabled with a HW Trigger. For example, software can set up EVT disabled then program a HW trigger to match an executed address at the beginning of a program to start it. Writing 1 will turn EVT on. Note that a change in value will generate a trace-on or trace-off Event Trace packet packet.	RW	0
2	trTeEvtTrigEn	1: Allows **trTeEvtTracing** to be set or cleared by Trace-on and Trace-off signals generated by the corresponding **Trigger Module**.	RW	0
4:3	trTeEvtCallMode	0: use messages that include both source and destination addresses for calls (CALL_SYNC, CALL_FULL). These generate more bytes (than modes 1, 2), however the decoder can determine the symbolic line # where calls are made and returned to without requiring a binary image. (Required if Call/Ret implemented) 1: Generate CALL_DEST for both direct and indirect calls. 2: least bandwidth: use CALL_DEST for indirect calls, CALL_PC for direct (inferrable) calls. Most aggressive on saving bandwidth 3: reserved	WARL	0
6:5	trTeEvtRetMode	0: use messages that include both source and destination addresses for returns (RET_FULL). These generate more bytes (than modes 1, 2), however the decoder can determine the symbolic line # where return is made and returned to without requiring a binary image. (Required if Call/Ret implemented) 1: generate RET_DEST for returns. 2: least bandwidth: use RET with no addresses when return address matches the call, otherwise RET_DEST. 3: reserved	WARL	0
7	Reserved			
8	trTeEvtExceptionEn	0: don't trace; 1: trace exceptions at the source.	WARL	0
9	trTeEvtInterruptEn	0: don't trace; 1: trace interrupts at the source.	WARL	0
10	trTeEvtCallRetEn	0: don't trace; 1: trace calls and returns	WARL	0

Bits	Field	Description	RW	Reset
11	trTeEvtSContextPrivEn	0: don't trace; 1: trace when Supervisor Context (CSR 0x5a8) is written to. Note: Context writes are also enabled when the **trTeControl.trTeContext** bit is set (only mentioned here).	WARL	0
12	trTeEvtHContextPrivEn	0: don't trace; 1: trace when Hypervisor Context (CSR 0x6a8) is written to.	WARL	0
13	trTeEvtTrapReturnEn	0: don't trace; 1: trace interrupts and trap returns (hardware does not distinguish exception and interrupt returns)	WARL	0
14	trTeEvtDbgTrigEn	0: don't trace; 1: trace debug trigger/watchpoints. The user sets up debugger hardware triggers with the Action value of 4 (Trace notify) that generates a trace packet when the trigger/watchpoint matches. This control bit enables or disables the generation of these trace messages.	WARL	0
15	trTeEvtUserDefinableEn	0: don't trace; 1: trace user definable signals. When any signal is asserted, a trace packet is generated that includes all the signal values at that moment.	WARL	0
16	trTeEvtPCSampEn	0: don't trace; 1: enable the tracing of periodic PC samples. The user sets up period of sampling and optionally the inclusion of randomizing the period.	WARL	0
17	trTeEvtExtTrigEn	0: don't trace; 1: enable the tracing of external trigger signals. The user sets up which external signals to trace and the edge detection (rising, falling, or both) for each signal.	WARL	0
18	trTeEvtPerfCtrSelEn	0: don't trace; 1: enable the tracing of performance counter selector outputs. The user first programs the mhpmeventX registers that generate the outputs to trace.	WARL	0
22:19	Reserved	Reserved for future Event Type enables		
23	trTeEvtPCSampRandEn	0: do not enable randomizing of sample period; 1: enable randomizing of sample period. The random count is added to the powers-of-2 count down set by the user. An LFSR is used to generate the random count. The width of the random count is user selectable (recommended at 6 bits), but fixed at design time.	WARL	0
27:24	trTeEvtSampDuration	Periodic sample duration for PC sampling events. Interval duration is exponential in powers of 2 - cycles = 2^(trTeEvtSamp Duration+6). i.e. range is 2^6 to 2^21 (64 to ~2 million cycles). For a 1GHz clock, the range is 64ns to 2ms.	WARL	0
28	trTeEvtLockout	0: ignore Trace Encoder FIFO-AF (almost-full) condition 1: when Trace Encoder FIFO-AF feedback signal indicates "almost full", stop generating events. This is a front-end throttle to prevent the Trace Encoder FIFO from filling up and using the Encoder back-end to stop tracing, which generates an error packet when tracing is resumed. When enabled, this lockout affects all external inputs to the encoder including Debug trigger/watchpoints, User definable Signals, External Signals (triggers), and Performance Counter selectors.	WARL	0
31:29	Reserved			

For the setup of **trTeEvtExceptionEn** *and* **trTeEvtInterruptEn**, *the return from an exception or interrupt are common and called a "trap return". Hardware tracing cannot*

distinguish between a return caused by an interrupt or an exception so if one or the other is turned off with these control bits, all trap returns will still be generated as EVT packets. The Trace decoder must deal with trap returns that may not have an associated exception or interrupt event. Privilege level filtering can be used to selectively quash some of them. The user may also choose to turn off the tracing of trap returns separately.

6.2.2. trTeEvtFeatures: Event Trace Features RO Register (trBaseEncoder+0x104)

Table 7. Event Trace Features Read-only Register - **trTeEvtFeatures** *(32b)*

Bits	Field	Description	RW	Reset
0	trTeEvtIntExcImpl	Read as 1 if Interrupt and Exception event types are implemented	RO	1 (SD)
1	trTeEvtCallRetImpl	Read as 1 if Call and Return eventtypes are implemented	RO	1 (SD)
2	trTeEvtContextPrivImpl	Read as 1 if Context and Privilege event types are implemented	RO	1 (SD)
3	trTeEvtDbgTrigImpl	Read as 1 if Debug Trigger event types are implemented	RO	1 (SD)
4	trTeEvtPCSampImpl	Read as 1 if PC Sampling is implemented	RO	1 (SD)
5	trTeEvtExtTrigImpl	Read as 1 if External Signal (Trigger) event types are implemented	RO	1 (SD)
6	trTeEvtPerfCtrSelImpl	Read as 1 if Performance Counter Selector event types are implemented	RO	1 (SD)
10:7	trTeEvtPPCsImpl	Number of Profiling Performance Counters implemented (0-15)	RO	8 (SD)
11	trTeEvtPPCtype	Type of Profiling Performance Counters implemented: 0: independent counters; 1: based on hpmcounters	RO	0 (SD)
12	trTeEvtPPCSelType	Type of Profiling Performance Counter selectors implemented: 0: independent selectors; 1: based on mhpmeventX	RO	0 (SD)
13	trTeEvtDbgTrigIDImpl	Read as 1 if Debug Trigger ID is implemented in RHTI (or manually)	RO	0 (SD)
16-14	trTeEvtSelectorsImpl	Number of Performance Counter selectors implemented for Event Trace (0-7)	RO	0 (SD)
31:17	Reserved			

SD = System Dependent. Reset value of a field is dependent on the build.

6.2.3. trTeEvtMissed: Missed Register (trBaseEncoder+0x108)

Event Missed. This register defines a 16 bit **Counter** which is incremented whenever an event is discarded because the Encoder cannot accept the event. The cause can be the encoder backend if full or the FIFO_AF (almost full, i.e. back pressure) signal is telling the event sourcing hardware to halt event generation until the backend has sufficient room to accept more events.

- The counter is 16 bits and is saturating.
- The counter can be read and cleared by software.
- Setting **trTeEvtControl.trTeEvtTracing**, i.e. starting event trace, will also clear this counter.

6.2.4. trTeLeafFilterControl: Leaf Function Filter Control Register (trBaseEncoder+0x10C)

Table 8. Event Trace Leaf Filtering Control Register - **trTeLeafFilterControl** *(32b)*

Bits	Field	Description	RW	Reset
3:0	trTeLeafFilterThreshold	Threshold for filtering out call-return pairs that are less than N cycles in duration. The threshold is set in powers-of-2, i.e. 2^N cycles, where N=0 to 15, a range of 2 to 16,384 cycles. A value of 0 disables leaf function filtering.	WARL	0

6.2.5. trTeEvtPPCEnables: Profiling Performance Counter (PPC) Enable Register (trBaseEncoder+0x110)

Each Event Type can be separately programmed to output PPC values when that Event Type is enabled. This register defines the enable bit for each Event Type to output PPC values in a packet immediately after the event.

The value of the field is always 0 if the feature is not implemented, thus by writing 1 to each field, the WARL return value indicates if the feature is implemented. The reset value is also 0 for all fields which means that by default no Event Types will output PPC values until the user enables them.

Table 9. Event Trace PPC Enables Register - **trTeEvtPPCEnables** *(32b)*

Bits	Field	Description	RW	Reset
0	trTeEvtExcPPC	0: don't trace PPCs; 1: trace PPCs for Exceptions	WARL	0
1	trTeEvtIntPPC	0: don't trace PPCs; 1: trace PPCs for Interrupts	WARL	0
2	trTeEvtCallRetPPC	0: don't trace PPCs; 1: trace PPCs for Calls and Returns	WARL	0
3	trTeEvtSContextPrivPPC	0: don't trace PPCs; 1: trace PPCs for Supervisor Context and Privilege mode changes	WARL	0
4	trTeEvtHContextPrivPPC	0: don't trace PPCs; 1: trace PPCs for Hypervisor Context and Privilege mode changes	WARL	0
5	trTeEvtTrapRetPPC	0: don't trace PPCs; 1: trace PPCs for Trap Returns	WARL	0
6	trTeEvtDbgTrigPPC	0: don't trace PPCs; 1: trace PPCs for Debug Trigger (Watchpoint) occurrences	WARL	0
7	trTeEvtPCSampPPC	0: don't trace PPCs; 1: trace PPCs for PC sampling events	WARL	0
8	trTeEvtExtSigPPC	0: don't trace PPCs; 1: trace PPCs for External Signal (Trigger) events	WARL	0
9	trTeEvtPerfCtrSelPPC	0: don't trace PPCs; 1: trace PPCs for Performance Counter Selector events	WARL	0
15:10	Reserved			
19:16	trTeEvtPPCNumTrace	0: turn off tracing of PPCs for all Event Types 1-15: Trace this number of PPCs. This tells the Encoder how many counters to include in the PPC trace packet. If the value is greater than the number of PPCs implemented, then trace all PPCs.	WARL	0
31:20	Reserved			

6.2.6. trTeEvtPPCControl: Profiling Performance Counter (PPC) Control Register (trBaseEncoder+0x114)

A write to the **trTeEvtPPCControl** register (any value) causes all the PPCs, i.e. **trTeEvtPPCCounter*i***, to be snapshotted (latched) allowing their respective values to be read by software. The PPCs are reset to zero and begin counting the pre-programmed events on the next cycle.

This action only works if Event Trace is disabled.

6.3. PPC Counter Registers - trTeEvtPPCCounter*i* (24b) (trBaseEncoder+0x150-0x198)

This set of registers is only applicable if the implementation includes independent PPC counters Section 4.2 or Section 4.3. The recommended width of each counter is 24 bits. They are delta, saturating counters that, for Event Trace, count performance events for the duration of each Event Trace event that is then packetized and sent into the trace stream. (Unfortunately the term "event" is an overloaded term in this context.)

This register access is an optional feature. If implemented, these registers are memory-mapped to allow for read/write access and that does not require cpu instructions to access them (as CSR-based HPM counters require).

The mapping of these registers to MMIO space is optional. The purposes are:

1. read/write access for DV-based testing, in-circuit diagnostic testing, and
2. to be used as independent (outside of Event Trace) perf counters. This extends the usefulness of these counters beyond Event Trace and HPM counters. For example, they can enable memory-based access by another core in the cluster or by JTAG or other external debug port/device for taking performance measurements without cpu intervention.

The registers exist in the design as Capture registers. When a new Event Trace event occurs, each of the profiling performance counters must be latched in order for the counter to be zeroed and ready to begin counting on the next cycle. Then the Trace Encoder must create a packet packet with each of the PPCs put into the packet, sequentially. This requires that all the PPCs be latched simultaneously.

If MMIO access is implemented, software can first snapshot the counters then read the Capture registers sequentially, until the next snapshot. The snapshot action is done with a write to a specific control register (s)

The **trTeEvtPPCCounter*i*** registers are similar to hpmcounterX counters of the HPM (Hardware Performance Monitor). This set of registers is based on the PPC Section 4.2 or Section 4.3 where the PPCs are independent Profiling Performance counters.

Table 10. Event Trace **trTeEvtPPCCounter*i*** *Capture Registers (24b)*

Addr Offset	Name	Size	RW	Reset	Description
0x120	PPCCounter0	[23:0]	RW	Undef	Counter 0 and 1 can be access as one 64b read with PPCCounter0 in the lower 32 bits and PPCCounter1 in the upper 32 bits.
0x124	PPCCounter1	[23:0]	RW	Undef	For 64b cores, this allows for more efficient reading of two counters with one read cycle.
0x128	PPCCounter2	[23:0]	RW	Undef	
0x12C	PPCCounter3	[23:0]	RW	Undef	
0x130	PPCCounter4	[23:0]	RW	Undef	
0x134	PPCCounter5	[23:0]	RW	Undef	
0x138	PPCCounter6	[23:0]	RW	Undef	
0x13C	PPCCounter7	[23:0]	RW	Undef	

6.4. PPC Event Selector Registers: trTeEvtPPCEvent*i* (64b) (trBaseEncoder+0x160-0x198)

The **trTeEvtPPCEventi** registers are equivalent to the **mhpmeventX** selectors of the HPM (Hardware Performance Monitor). This set of registers is based on the PPC Section 4.2 where the PPCs and event selectors are independent.

The Event Trace developer has the choice of using mhpmeventX selectors (components of the HPM - Hardware Performance Monitor) or implementing PPCEvent*i* selectors which are independent of the HPM. Each PPCEvent*i* selector register is 64 bits and is identical programmatically to mhpmeventX selectors of the HPM (with one exception that the interrupt bit 63 status/control bit is undefined). These registers can be read/written as 64-bit registers or as address-adjacent 32-bit low/high registers.

This Event Trace specification and associated memory mapping is assuming the maximum number of PPCs is 8. Thus the *i* in the register name is an index from 0 to 7. If the implementation includes fewer than 8 PPCs, then the unused register slots are reserved.

Table 11. Event Trace PPC Event Selector Register **trTeEvtPPCEventi** *fields*

Bits	Field	Description	RW	Reset
[31:0]	Low	Lower 32 bits of PPC event selector. (address 0x160, 0x168, 0x170, 0x178, 0x180, 0x188, 0x190, or 0x198)	RW	0
[63:32]	High	Upper 32 bits of PPC event selector. For XLEN=32, access is through the address of trTeEvtPPCEvent*i* + 4. (0x164, 0x16C, 0x174, 0x17C, 0x184, 0x18C, 0x194, or 0x19C)	RW	0

6.4.1. trTeEvtExtSigControl: External Signal Control Register (trBaseEncoder+0x1A0)

Event Trace supports up to 8 external signals that can be traced as events. The user programs the edge detection for each signal (don't trace, rising, falling, or both) in the **trTeEvtExtSigEnables** register. When the corresponding signal edge is detected, an Event Trace packet is generated with the signal ID and the PPC values, if enabled.

These fields are WARL; the value of the field is always 0 if the feature is not implemented, thus by writing 1, 2, or 3 to each field, the WARL return value indicates if the feature is implemented. The reset value is also 0 for all fields which means that by default no external signals will be traced until the user enables them.

Table 12. Event Trace External Signal Control - **trTeEvtExtSigControl** *(32b)*

Bits	Field	Description	RW	Reset
1:0	trTeEvtExtSig0Edge	0: don't trace; 1: trace rising edge 2: trace falling edge 3: trace both edges	WARL	0
3:2	trTeEvtExtSig1Edge	0: don't trace; 1: trace rising edge 2: trace falling edge 3: trace both edges	WARL	0
5:4	trTeEvtExtSig2Edge	0: don't trace; 1: trace rising edge 2: trace falling edge 3: trace both edges	WARL	0
7:6	trTeEvtExtSig3Edge	0: don't trace; 1: trace rising edge 2: trace falling edge 3: trace both edges	WARL	0
9:8	trTeEvtExtSig4Edge	0: don't trace; 1: trace rising edge 2: trace falling edge 3: trace both edges	WARL	0

Bits	Field	Description	RW	Reset
11:10	trTeEvtExtSig5Edge	0: don't trace; 1: trace rising edge 2: trace falling edge 3: trace both edges	WARL	0
13:12	trTeEvtExtSig6Edge	0: don't trace; 1: trace rising edge 2: trace falling edge 3: trace both edges	WARL	0
15:14	trTeEvtExtSig7Edge	0: don't trace; 1: trace rising edge 2: trace falling edge 3: trace both edges	WARL	0
21:16	trTeEvtExtSigMux	Control field to select a developer-provided multiplexer of N groups of 8 signals (where N can be 1 to 64) external signal groups to trace. For example, if the developer has 32 external signals they want to be able to trace, the developer implements a mux with 4 input sets of 8 signals per set.	WARL	0
22	trTeEvtExtSigPCMode	0: Do not include the PC in the external signal trace packet (to save on bandwidth) 1: Include PC in external signal trace packet, to correlate transition of signal to instruction executed in core	WARL	0
23	Reserved			
27:24	trTeEvtExtSigInterval	The **cycles-interval** is the equivalent of a "one-shot" circuit, triggered by the (user-selectable) signal edge. While the interval is true, any additional edges causes the recording state to change from "edge" to "multi-edge". The cycles-interval is a power-of-2 number of cycles, i.e. 2^(<value>) with a range from 0 (no interval), 2, 4, 8, ... 16,384, 32,768 cycles. This field is common for all signals.	WARL	0
31:28	Reserved			

6.4.2. trTeEvtPerfSelectControl: Performance Selector Control Register (trBaseEncoder+0x1A4)

The Event Trace Performance Selector feature supports the conversion of a HPM (Hardware Performance Monitor) selector (mhpmeventX) output assertion into an Event Trace packet, and records the number of additional times this happens in a user-selectable interval. In other words, the signal from an mhpmeventX routed to and counted by a standard programmable performance counter is also routed to the Trace Encoder to be recognized and recorded, with timestamp, in the trace timeline.

Performance counters count these occurrences; this Event Trace feature does much more - it records the instruction at the time of the occurrence and the timestamp creates a time history of these occurrences. In addition, if the event occurs frequently, the hardware compresses these into a "first occurrence" and count, recording the number of times the event occurs within the user-specified interval that can be visualized as frequency of occurrence over a time history. The purpose of the interval is to reduce the number of events recorded when the event occurs frequently, for example, counting load instructions.

The start of the event requires a rising edge on the Select signal but the counting of the event during the interval is done by counting the level (high) of the signal.

The logic for this event type requires a counter to count the number of cycles the signal is high during the interval. The counter is saturating (stops counting when reaching maximum value). The recommended width is 24 bits and leading-zero suppression for recording the value in the trace packet.

The trace packet requires that the PC (address of latest retired instruction) and timestamp be latched at first occurrence of the selector event edge; the counter counts the high level during

the interval, then the packet combines the PC, event count, and starting timestamp when the interval ends.

The counter is not accessible as a memory-mapped register. It is part of the Perf Selector Event Trace logic and the value is stored in the packet and reset to zero at the end of the interval. For reasons of access for DV, it may be of interest to make it read/write-able as a memory-mapped register, but that is not part of this specification.

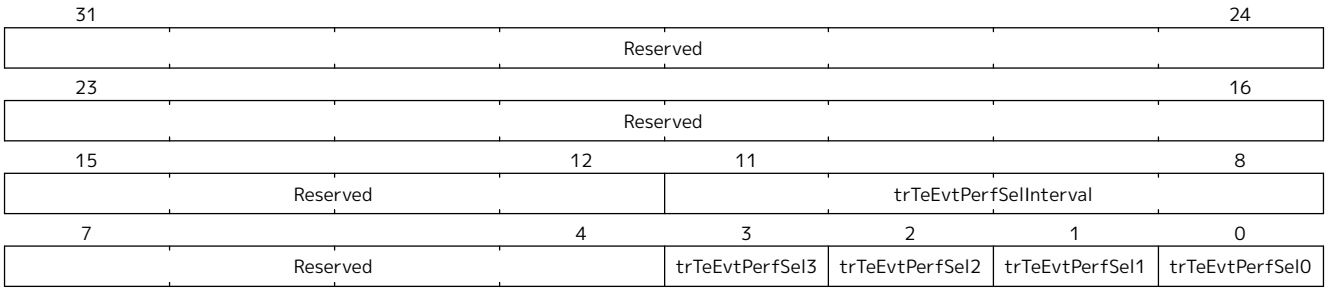


Figure 2. Event Trace Performance Selector Control - **trTeEvtPerfSelectControl** *(32b)*

Table 13. Event Trace Performance Selector Control - **trTeEvtPerfSelectControl** *(32b)*

Bits	Field	Description	RW	Reset
0	trTeEvtPerfSel0	0: don't trace; 1: trace rising edge of selector output; count level (high) during time interval	WARL	0
1	trTeEvtPerfSel1	0: don't trace; 1: trace rising edge of selector output; count level (high) during time interval	WARL	0
2	trTeEvtPerfSel2	0: don't trace; 1: trace rising edge of selector output; count level (high) during time interval	WARL	0
3	trTeEvtPerfSel3	0: don't trace; 1: trace rising edge of selector output; count level (high) during time interval	WARL	0
7:4	Reserved			
11:8	trTeEvtPerfSelInterval	**cycles-interval** during which perf selector level (high) events are counted. Interval duration is selectable in powers of 2 **cycles-interval = 2^(trTeEvtPerfSelInterval + 6)** Range is 2^6 to 2^21, i.e. 64 cycles to ~2 million cycles in powers-of-2 values. For a 1GHz clock, the range is 64ns to 2ms. Common for all Perf Selects.	WARL	0
31:12	Reserved			

Chapter 7. RHTI Extensions for Event Trace

This file (RHTI-Extensions.adoc) defines extensions to the "RISC-V Hart to Trace Interface Specification" [1] to support several features for Event Trace. While an implementer can wire up these signals "on their own", RHTI defines them, along with all the existing signal definitions for Instruction Trace, to provide the named signals for Event Trace use.

7.1. Trigger/Watchpoint ID

As explained in the RHTI [1] section of the specification, the debug module of the RISC-V hart may have a "Trigger Unit", also named "Trigger Module" in the RISC-V Debug specification [2]. A 4-bit action field defines what happens when a trigger/watchpoint match occurs. Encoded codes 2-5 are for trace use. The value 4 is **Trace-notify**.

The RHTI defines trigger [2] signal to be **Trace-notify**. When asserted, it indicates a match on one of the programmed hw triggers.

The following signal definition extends the section "Optional sideband signals", Table 8 of the spec with the definition of **triggerID** which identifies **which** trigger matched.

Table 14. Optional sideband encoder **triggerID** *output signals*

Signal	Group	Function
triggerID	OR ("Optional Replicated" which means replicated N time for a hart that can retire N instructions per cycle with a trigger match on any of those instructions)	4-bit field that encodes the ID of the matching trigger, thus supports up to 16 triggers. There are two cases for multiple triggers matching on the same cycle: 1) multiple triggers with the same match condition, which could include address-range triggers that overlap. (Also there is nothing to prevent a user from programming multiple triggers with the same match condition), or 2) multiple triggers that occur when multiple instructions retire in the same cycle. The Event Trace debug trigger/watchpoint event is limited to one trigger match per cycle. In the case of multiple matches, for Case 1, the RHTI hardware must use numeric precedence to transmit (over this TriggerID field) the lowest ID value. For Case 2, each of the potential matches on each of the retired instructions has its own, internal, 4-bit field. Logic in the core ORes the occurrence bits and also uses precedence to select the lowest numbered triggerID to transmit over the RHTI signal field.

7.2. HPMeventX Output

Table 15. Optional sideband event selector outputs **HPMEventX** *for Event Trace Encoder*

Signal	Group	Function

mhpmeventXOutput[*eventNum_p-1*:0]	O	Field that carries (routes) the least significant "Num" Hardware Performance Monitor (HPM) mhpmEventX selector outputs, where Num is *eventNum_p*. These signal outputs may be used for profiling performance counter delta counting or Selector-based event generation or both. If HPM mhpmeventX selectors are being used (Alternative 2), *eventNum_p* can range from 1 to 8.

Instruction Retired Count

Table 16. Optional sideband encoder **InstRetCount** *output signals*

Signal	Group	Function
instretcount[*countwidth_p-1*:0]	O	Number of instructions retired in this cycle. This is usable by trace to accumulate the total number of instructions retired at the time of a trace message. The width of this field is *countwidth_p* which can be as small as 1 and as large as the encoded max number of retired instructions in any one cycle. For example, if a core can retire up to 4 instructions per cycle, then a 3-bit width is required to encode the count of retired instructions from 0 to 4.

Chapter 8. Trace Control Extensions for Event Trace

This file (Trace-Control-Extensions.adoc) defines the extensions required to the "RISC-V-Trace-Control-Interface.pdf" spec to support Event Trace.

The Trace Control Interface spec, Table 4, has several reserved address ranges labeled "Reserved for future standard extension". Event Trace control registers starting point is an offset from the Trace Encoder base address. The Event Trace allocates and uses addresses **0x100-0x17F**. Thus **trTeEventBase** = **trBaseEncoder+0x100**.

8.1. Event Trace Control Register Mode Field

This document defines Event Trace as a sub-component of the Trace Encoder. Event Trace as a trace mode (separate from Instruction Trace) is set by software writing

trTeControl.trTeInstMode = 4

which is defined in the RISC-V Trace Control Interface spec as **"Protocol defined trace mode"** [3 pp. Trace Encoder Control Interface].

8.2. Time Synchronization Message for simultaneously recording mtime & timestamp

This message is common for N-Trace and E-Trace Instruction Trace and its purpose is to record mtime and trace timestamp simultaneously for Trace message synchronization. This message is generated periodically based on a configurable interval of N- or E-Trace messages, and also aperiodically after certain events such as trace-on, exit debug, exit reset, ProgTraceSync message, and immediately after trace has stopped or halted.

MTIME_SYNC Periodic Generation The period for generating MTIME_SYNC is defined by a field in the register **trTsControl** inside the TimeStamp (TS) sub-component of Instruction Trace. The field, [14:10], of the ratified spec is Reserved. The lower 4 bits are, in this spec, being relabeled **trTsControl.trTsSyncMax**. This controls the interval for generating the MTIME_SYNC message based on a count of the number of N-Trace messages output.

Interval Count The interval count is 2^(7+trTsSyncMax) where **trTsSyncMax** usable values ranges from 1 to 15, where 0 is **"off"**. This provides a range from 2^8 (256) to 2^22 (~4x10^6). Note that this field is WARL so some of the upper values could be left out to reduce hardware.

8.2.1. Register trTsControl: Timestamp Control Register Field

*Table 17. Timestamp Control Register Field - **trTsControl.TrTsSyncMax** (32b)*

Bits	Field	Description	RW	Reset
13:10	trTsSyncMax	**0**: MTIME_SYNC disabled **1-15**: Period for generating MTIME_SYNC message is 2^(7+trTsSyncMax) N-Trace messages	WARL	0

Engineering (non-normative) NOTE: MTIME_SYNC can be implemented by using a 16-input multiplexer; the first input selects always-0, the remaining 15 inputs connect to the appropriate divide-by-

2 taps of the binary counter of N-Trace messages, starting at bit 7 and ending at bit 22, with the 4 bits of trTsSyncMax as the mux selector. This logic then generates an output pulse of MTIME_SYNC at the rising edge of the mux output. As mentioned, a narrower mux could be used in the actual implementation to save on hardware. A 10-input mux would have a max period of 2^17, or 128K of N-Trace messages.

8.3. MTIME_SYNC Aperiodic Generation

If enabled, an MTIME_SYNC message shall be generated sometime after (when is not critical):

1. a trace-on, exit debug, or exit reset message
2. immediately after trace has stopped or halted (this needs to be timely)

This provides the time synchronization after a gap in trace and at the end of the trace buffer. The interval timer must also be reset.

Chapter 9. N-Trace Message Definitions

the N-trace specification uses the term "message" to refer to the data output by the trace encoder thus this chapter will also use that term. Other documents use the term "packet" so the other chapters of this document will use the term "packet" to refer to the data output by the trace encoder. The terms are interchangeable and both refer to the same thing.

This file (N-Trace-packets.adoc) defines the Event Trace messages that extend N-Trace [5].

9.1. N-Trace TCODE Summary

Value	Description
0x22 (34) (compressed), 0x23 (35) (synchronized)	EVT Trace Messages for non-function events * EVTSRC=0: Sync Enable * EVTSRC=1: Sync Disable * EVTSRC=2: Exception * EVTSRC=3: Interrupt * EVTSRC=4: Return from trap * EVTSRC=5: Debug Trigger/Watchpoint * EVTSRC=6: Periodic PC Sampling * EVTSRC=7: External Signal (trigger) * EVTSRC=8: Perf Counter Selector Occurrence
0x24 (36)	CALL_SYNC Function Event
0x25 (37)	CALL_FULL Function Event
0x26 (38)	CALL_DEST Function Event
0x27 (39)	CALL_PC Function Event
0x28 (40)	RET_FULL Return Event
0x29 (41)	RET_DEST Return Event
0x2A (42)	RET Function Event
0x11 (17)	Profiling Performance Counter Recording
0x2 (2)	Ownership: Process Information (Context ID and Privilege Mode)
0x10 (16)	N-Trace MTIME_SYNC Message for Timing Synchronization

9.2. Informational (non-normative)

- **SRC** width is a hardware build-time configuration based on the total number of harts, defined in the Trace Control Interface Spec as trTeSrcBits
- **TSTAMP** is the N-Trace accumulating, running timestamp. Its clock is required to provide accurate short-duration execution timing, with a 10ns minimum resolution and a preferred 1ns resolution (100MHz - 1GHz). TSTAMP minimum width should be 40 bits which provides 18 minutes of duration at 1GHz clock rate. If Event Trace is to run for hours, then 48 bits provides 78 hours.
- **cfg** in the Bits column means the number of bits is configurable at build time. The primary example is the number of bits for the SRC field. This represents the total number of Harts in the SoC.
- **N-Trace** uses a byte-sized data format where 6 bits are payload and 2 bits called MSEO bits define the message framing. 00 = first or middle bytes, 01 = end of a variable field, 11 = end of a message/packet.

- F-ADDR means the full address is included in the message. U-ADDR means the unique part of the address is included in the message. Both are output with leading zero suppression.

9.3. Non-Function Event Trace Message Formats

Event Trace has defined a new N-Trace message called EVT that covers non-call/ret event types. There are two types: EVT compressed (TCODE=34) and synchronized (TCODE=35). Note that this message type has the same TCODEs and format as the Nexus Trace ICT (in-circuit trace) messages.

Table 18. EVT Trace Messages for non-function events

Bits	Field	Description
6	TCODE	Value = 0x22 (34) (compressed) or 0x23 (35) (synchronized)
Cfg	SRC	Source Hart of this message (width is trTeSrcBits)
4	EVTSRC	Type of EVT message; 4 bits provides up to 16 types
2	EVTCNT	0=One field (EVTPC), 1=Two fields (EVTPC, EVTDATA), 2-3=reserved Note: EVTSRC+EVTCNT equal 1 byte of N-TRACE message output
var	EVTPC	(TCODE=0x22) U-ADDR, (TCODE=0x23) F-ADDR. Source address for some EVTSRC types
var	EVTDATA	Additional message content. Some values are subtypes. Some are an additional address. For addresses, (TCODE=0x22) U-ADDR, (TCODE=0x23) U-ADDR (since first addr is F-ADDR)
var,cfg	TSTAMP	(TCODE=0x22) Relative timestamp value (TCODE=0x23) Absolute timestamp value

9.3.1. Event Trace non-Function Messages, per-EVTSRC (type)

Table 19. Event Trace Message Fields for non-function events

Event	EVTSRC	EVTCNT	EVTPC	EVTDATA	Description
Sync Enable	0	1	PC	SYNC	EVTPC is always F-ADDR (TCODE=0x23) PC=instruction responsible for trace-on. SYNC field, must be compatible w/N-Trace Table 25: (1) = Exit from Reset (3) = exit-debug. PC=first instruction to execute after debug mode. (5) = event trace enable. (others) Reserved Note: some cores may not have logic to recognize "Exit from Reset". Start of execution can use the "event trace enable" encoding.
Sync Disable	1	1	PC	EVCODE	EVCODE field (compatible w/N-Trace Table 24): EVTPC is U-ADDR (TCODE=0x23) (0) = enter-debug. PC=address loaded into depc (has not executed). (4) = event trace disabled. PC=instruction responsible for trace-off. (6) = enter-reset. PC=last instruction to execute before reset. (others) reserved

Event	EVTSRC	EVTCNT	EVTPC	EVTDATA	Description
Exception	2	1	PC	cause	(ITYPE=1) PC is what is loaded into xepc (the address of the instruction that caused the exception), cause is what is loaded into the Exception Code field of xcause. cause signals are defined in Table 1 of RHTI spec (see References).
Interrupt	3	1	PC	cause	(ITYPE=2) PC is what is loaded into xepc, cause is what is loaded into the Exception Code field of xcause.
Return from trap	4	1	PC	PCdest	(ITYPE=3) Return from exception or interrupt (MRET or SRET) (to distinguish from normal function return) This is generated when both PC's are not filtered by privilege mode.
Return from trap	4	0	PCdest	---	(Variant of above) Return from exception or interrupt (MRET or SRET) Generated when the source PC is filtered by privilege mode (but PCdest is not).
Debug Trigger/ Watchpoint w/TRIG_ID	5	1	PC	TRIG_ID	When a debug trigger/watchpoint matches, the PC is normally the address of the instruction that caused the match, although for a watchpoint, it can be after when comparing on the data portion of a load or store. In multi-issue cores, this message may report the address of a later instruction retired in the same cycle. For **trTeEvtFeatures.trTeEvtDbgTrigIDImpl** [DebugTriggerIDImpl] = 1, the **TRIG_ID** is the trigger number and is recorded in the EVTDATA field.
Debug Trigger/ Watchpoint wo/TRIG_ID	5	0	PC	---	When a debug trigger/watchpoint matches, the PC is normally the address of the instruction that caused the match, although for a watchpoint, it can be after when comparing on the data portion of a load or store. In multi-issue cores, this message may report the address of a later instruction retired in the same cycle. For **trTeEvtFeatures.trTeEvtDbgTrigIDImpl** [DebugTriggerIDImpl] = 0, the hardware does not provide the trigger ID; the trace decoder must determine the unique trigger indentification from the PC and from the knowledge of the trigger values that are programmed.
User Definable Signals	6	0	PC	impdef[] signals value	This event samples the PC address of the last instruction retired at the time of the assertion of one of the impdef[] signals [1 pp. Optional sideband signals].
Periodic PC Sampling	7	0	PC	---	This event samples the PC of the hart at a periodic interval. The PC is the address of the last instruction retired at the time of the sample.
External Signal (trigger) with PC	8	1	PC	ES_VAL	**ES_VAL**[4:0] = {SIG_ID[2:0],SIG_VAL[1:0]} where SIG_VAL is rising (0), falling (1), both (2), multiple edges (3). Generated when external signal edge on this channel (SIG_ID) is detected by the Event Trace encoder. The PC is the address of the closest red instruction when the signal is recognized. Timestamp must be enabled, and its value is latched at the occurrence of the signal edge. **trTeEvtExtSigControl.trTeEvtExtSigPCMode** = 1 [ExtSigPCMode] to include the PC in the message.

Event	EVTSRC	EVTCNT	EVTPC	EVTDATA	Description
External Signal (trigger) no PC	8	0	ES_VAL	---	**ES_VAL** (see above). Generated when external signal edge on this channel (SIG_ID) is detected by the Event Trace encoder. Timestamp must be enabled, and its value is latched at the time of the signal edge. **trTeEvtExtSigControl.trTeEvtExtSigPCMode** = 0 [ExtSigPCMode] for no-PC mode.
Perf Counter Selector Occurrence	9	1	SEL_ID	Count	**SEL_ID** is the Selector (channel) recorded. **Count** is the number of cycles the signal has a high value (i.e. level sensitive) during the interval window, defined by the control register **trTeEvtControl.trTeEvtPerfSelInterval** [select-cycles-interval].

9.4. Event Trace Message Formats for Calls and Returns

The following table lists the message types to support function Calls and Returns, with different types and numbers of parameters. These are defined separately from the non-functional EVT messages to provide the most compact messaging.

If only the destination address of an event is recorded in the message, the decoder must determine the source address from the execution binary and the previous messages recorded. In some cases, it will not be possible to determine the exact source address when a function is called in many different places in the calling function. If the user wants the precise source and destination addresses then they must set the appropriate Event Trace modes (and there are separate modes for calls and returns), with the understanding that these messages include more bytes.

Table 20. Event Trace Message Types for Calls and Returns

Name	TCODE	Source	Destination	ITYPE	Description
CALL_SYNC	36	PC (F-ADDR)	Dest Addr (U-ADDR)	8 or 9	Traces both Src and Dest addresses. The second address is XOR compressed. Full (absolute) timestamp if enabled.
CALL_FULL	37	PC (U-ADDR)	Dest Addr (U-ADDR)	8 or 9	Traces both Src and Dest compressed (unique) addrs; relative timestamp if enabled.
CALL_DEST	38	-----	Dest Addr (U-ADDR)	8 or 9	itype=8 means indirect (uninferrable) call or co-routine swap. Source function can be derived from trace history however Src address (i.e. line # of caller) is unknown.
CALL_PC	39	PC (U-ADDR)	-----	9	inferrable call so destination obtained from opcode if binary available
RET_FULL	40	PC (U-ADDR)	Dest Addr (U-ADDR)	13	Traces both Src and Dest (return to) addresses
RET_DEST	41		Dest Addr (U-ADDR)	13	return-to address; caller (function) determined from symbolic address range of destination address and/or from history.

Name	TCODE	Source	Destination	ITYPE	Description
RET	42	------	------	13	Function returned; the return destination is not important.

Recommended priority of implementation
1: CALL_SYNC, CALL_FULL, RET_FULL
2: CALL_PC, RET
3: CALL_DEST, RET_DEST

9.4.1. CALL_SYNC Function Event

Used to synchronize messages, outputting new absolute address for any call type (8, 9)

Table 21. CALL_SYNC Function Event

Bits	Field	Description
6	TCODE	Value = 0x24 (36)
cfg	SRC	Source Hart of this message (width is trTeSrcBits)
var	Src Addr	PC: F-ADDR (F=full)
var	Dest Addr	U-ADDR
var,cfg	TSTAMP	Absolute timestamp value

9.4.2. CALL_FULL Function Event

Trace both source and destination addresses of any call type (8, 9)

Table 22. CALL_FULL Function Event

Bits	Field	Description
6	TCODE	Value = 0x25 (37)
cfg	SRC	Source Hart of this message (width is trTeSrcBits)
var	Src Addr	PC: U-ADDR (U=unique part of address)
var	Dest Addr	U-ADDR
var,cfg	TSTAMP	Relative timestamp value (offset from last reported absolute or relative timestamp)

9.4.3. CALL_DEST Function Event

For Uninferrable call

Table 23. CALL_DEST Function Event

Bits	Field	Description
6	TCODE	Value = 0x26 (38)
cfg	SRC	Source Hart of this message (width is trTeSrcBits)
var	Dest Addr	U-ADDR
var,cfg	TSTAMP	Relative timestamp value

9.4.4. CALL_PC Function Event

For Inferrable call

Table 24. CALL_PC Function Event

Bits	Field	Description
6	TCODE	Value = 0x27 (39)
cfg	SRC	Source Hart of this message (width is trTeSrcBits)
var	PC	U-ADDR (itype=9; destination is inferrable from opcode)
var,cfg	TSTAMP	Relative timestamp value

9.4.5. RET_FULL Return Event

Trace both source and destination addresses of return when caller is known.

Table 25. RET_FULL Return Event

Bits	Field	Description
6	TCODE	Value = 0x28 (40)
cfg	SRC	Source Hart of this message (width is trTeSrcBits)
var	PC Addr	U-ADDR
var	Dest Addr	U-ADDR
var,cfg	TSTAMP	Relative timestamp value

9.4.6. RET_DEST Return Event

Output destination address only, to reduce bandwidth.

Table 26. RET_DEST Return Event

Bits	Field	Description
6	TCODE	Value = 0x29 (41)
cfg	SRC	Source Hart of this message (width is teImpl.nSrcBits)
var	Dest Addr	U-ADDR of destination address
var,cfg	TSTAMP	Relative timestamp value

9.4.7. Function RET Event

Trace hardware can maintain a one-level stack of the last caller PC address (+2 or +4) which is compared to the return destination address. If a match, the return matches the call so addresses are not needed for decode. This is the most compact message since no source or destination address is required.

Table 27. RET Function Event

Bits	Field	Description
6	TCODE	Value = 0x2A (42)
cfg	SRC	Source Hart of this message (width is trTeSrcBits)
var,cfg	TSTAMP	Relative timestamp value

9.5. Periodic Synchronization Messages

Messages that include the Program Counter (PC) are predominantly U-ADDRes. Periodically, however, the full address (F-ADDR) must be output to provide a synchronization point for the trace decoder. The period for generating a synchronization message is defined by the N-Trace register **trTeControl.trTeInstSyncMax** [3 pp. Trace Encoder Control Interface]. The count is 2^(trTeInstSyncMax+4).

Hardware counts the number of Event Trace messages that include a PC and outputs a synchronization message when the count reaches the programmed value. Messages that can include an F-ADDR include all EVT (TCODE=0x23) messages that include a PC address and CALL_SYNC.

9.6. Profiling Performance Counter (PPC) Recording

This message is used to record the Event Trace Profiling Performance Counter (PPCs) which are delta values. Since N-Trace variable fields use leading-zero suppression to reduce bandwidth, the number of bytes used to record each counter in many cases is reduced. The closer together events are in time, the smaller the delta count, thus fewer bits wide. Each byte carries 6 bits of counter value so the incremental sizes are 6 bits/1 byte, 12 bits/2 bytes, 18 bits/3 bytes, and 24 bits/4 bytes.

This message is generally output immediately after an event message when PPCs are enabled for that message. Thus it does not include a timestamp since the event message itself includes a timestamp, if enabled.

The number of PPCs implemented is defined in the control register **trTeEvtFeatures.trTeEvtPPCsImpl** [PPCsImplemented]. The number of PPCs to actively record per message is defined by the control register **trTePPCEnables.trTeEvtPPCNumTrace** [PPCsToTrace]. (Hardware must make sure that the number of PPCs to record does not exceed the number of PPCs implemented.) Thus this message can output a variable number of counters. The N-Trace message format defines the end of a message. [5 pp. Ch. MSEO Sequences]

Table 28. Profiling Performance Counter Recording

Bits	Field	Description
6	TCODE	Value = 0x11 (17)
cfg	SRC	Source Hart of this message (width is trTeSrcBits)
var	PPC0	Profiling Performance Counter0
var	PPC1	Profiling Performance Counter1 (if included)
...	...	...
var	PPC7	Profiling Performance Counter7 (if included)

9.7. Ownership N-Trace Message

N-Trace **Ownership** *Message [5 pp. Ownership Message] is already defined in the ratified N-Trace spec. It is here as reference to the message for tracing context ID and Privilege Mode as a selectable event type to trace, or when privilege mode is changed. This is described in detail in the ratified RISC-V N-Trace spec [5 pp. Ownership Message].*

Table 29. Ownership Message: Process Information (Context ID and Privilege Mode)

Bits	Field	Description
6	TCODE	Value = 0x2 (2)
cfg	SRC	Source Hart of this message (width is trTeSrcBits)
var	PROCESS	This is a variable-length field, which encodes V and PRV privilege mode bits as well as scontext/hcontext CSR values. PROCESS[x+5:0] = {CONTEXT[x:0], V[0], PRV[1:0], FORMAT[1:0]}
var,cfg	TSTAMP	Relative timestamp value

9.8. Time Synchronization message for recording mtime & timestamp

N-Trace MTIME_SYNC Message to record mtime and trace timestamp simultaneously for Timing Synchronization.

Table 30. N-Trace MTIME_SYNC Message for Timing Synchronization

Bits	Field	Description
6	TCODE	Value = 0x0F (15)
Cfg	SRC	Source Hart of this message (width is trTeSrcBits)
var	mtime	F (Full) mtime value
var,cfg	TSTAMP	F (Full) timestamp value

9.9. Inhibiting Trace for Security and Unbounded Wait reasons

The RHTI spec cite:[riscv-RHTI-spec-0.83, Trace State] added Trace State signal group **trcstate[1:0]** to replace the **halted** signal to provide additional conditions. Extended values are:

10 Inhibiting trace for security reasons
11 Unbounded wait - pausing trace for an unbounded amount of time (e.g. executing WFI or WFE instructions)

These states, when they occur over the RHTI, generate the N-trace ProgTraceCorrelation Message which is emitted when the trace is disabled or stopped. These additions are to the EVCODE field:

5 = Trace disabled for security reasons
6 = Trace paused for an unbounded amount of time

In addition, the **CDF** field includes a new value of **2**, for Event Trace, whose value means "do not generate the I-CNT or the HIST fields."

Chapter 10. E-Trace Packets

For Event Trace, E-trace encoders will generate packets closely related to the format 1 and 3 instruction trace packets.

Format 3 sync-event or sync-trap packets will be generated whenever a full address is required, otherwise format 1 packets will be used.

Although the packets are classified as format 3 and format 1, their payloads are not identical to their instruction trace equivalents, as event trace packets need to provide different information. The decoder can determine whether to decode using the insturction trace or event trace formats from the format 3 subformat 3 (support) packet which will be ouptut before the first traced instruction is reported.

10.1. Format 3 packets

Format 3 packets are used for synchronization, traps, reporting context and supporting information. There are 4 sub-formats.

10.2. Format 3 subformat 3 - Support

The structure of this packet is identical to the instruction trace sync-support packet as defined by the Standard Support Extension to E-trace. However, the **encoder_mode** field will be set to 1 to indicate Event Trace.

10.3. Format 3 subformat 0 - Event

This packet format is used whenever a full address is required, and the event being reported is not a trap. This includes privilege changes resulting from trap returns, as well as precise and asynchronous discontinuity context changes (see RHTI **ctype** signal).

The 2 reserved bits between **cause** and **address** keeps the address aligned at the same position in the packet as in the sync-trap packet. This eliminates any additional muxing cost that would otherwise arise by having to place the address at a different offset in this packet compared to sync-trap or sync-start packets.

Table 31. Packet format 3, subformat 1

Field name	Bits	Description
format	2	11 (sync): synchronisation
subformat	2	00 (event): Start of tracing, resync, or any time a full address is required
notify	1	Indicates if the packet was generated by a trace-notify trigger when 1.
privilege	*privilege_width_p*	The privilege level of the reported instruction.
time	*time_width_p*, or 0 if *notime_p* is 1	The time value.
context	*context_width_p*, or 0 if *nocontext_p* is 1	The instruction context
cause	4	If **notify** = 0, this field suppies the *itype* of the instruction. If **notify** = 1, it supplies the trigger ID.
reserved	*ecause_width_p* - 2	reserved

Field name	Bits	Description
address	*iaddress_width_p* - *iaddress_lsb_p*	Full instruction address. Address alignment is determined by *iaddress_lsb_p*: **address** must be left shifted by _iaddress-lsb_p_ in order to recreate original byte address.

10.4. Format 3 subformat 1 - Trap

This packet is identical to the instruction trace sync-trap packet, except that the **branch** bit has no purpose and can be considered redefined as 'reserved'.

10.5. Format 3 subformat 2 - Context

This packet is identical to the instruction trace sync-context packet, and is used to report imprecise context changes.

This file (E-Trace-packets.adoc) defines the E-Trace [6] packets for Event Trace.

10.6. Format 1 packets

This packet format is used for function calls and returns and notify triggers, whenever the address can be reported differentially.

Structurally it is arranged so that the **address** field starts at the same bit position as in the instruction trace format 1 packet reporting 1 branch.

The address is omitted for implicit returns.

Table 32. Packet format 1 - cause, address

Field name	Bits	Description
format	2	01 (diff-delta): includes event information and differential address
cause	4	If **notify** = 0, this field suppies the *itype* of the instruction. If **notify** = 1, it supplies the trigger ID.
iret	1	Always 0 in this format
src	1	When set to 1 indicates that **cause** is the itype of the call/return and **address** is the address of the target. **cause** field
address	*iaddress_width_p* - *iaddress_lsb_p*	Differential instruction address.
notify	1	If the value of this bit is different from the MSB of **address**, it indicates that this packet is reporting an instruction because of a trace-trigger.
updiscon	1	If the value of this bit is different from the MSB of **notify**, it indicates that this packet is reporting the instruction following an uninferable discontinuity and is also the instruction before an exception, privilege change or resync (i.e. it will be followed immediately by a format 3 *te_inst*).

Table 33. Packet format 1 - cause only

Field name	Bits	Description
format	2	01 (diff-delta): includes event information and differential address
cause	4	Supplies the *itype* of the return instruction.

Field name	Bits	Description
iret	1	Always 1 in this format

10.6.1. Format 1 src field

Call/return instructions and their respective targets are usually reported as separate consecutive packets. These packets will usually contain the **itype** corresponding **address**. However, if for some reason the call/return instruction is not reported (e.g. due to filtering) then the packet reporting the target will include the **itype** of the call/return along with the **address** of the target. This is indicated by the **src** field being 1.

10.7. Format 2 packets

This packet format is used for PC sampling.

Structurally it is identical to the instruction trace format 2 packet, but with all the bits above the address removed.

Table 34. Packet format 2

Field name	Bits	Description
format	2	10 (addr-only): differential address only
address	*iaddress_width_p* - *iaddress_lsb_p*	Differential instruction address.

10.8. Format 0 packets

Plan to define for Performance counters.

Appendix A: Event Trace Operations (non-normative)

- Event Trace is a mode of the Trace Encoder which is set up with **trTeControl.trTeInstMode=4**. Then, **trTeControl.trTeEnable** is used to enable trace or disable and flush it, with **trTeControl.trTeEmpty** used to indicate when the flush has completed.
- **trTeEvtControl.trTeEvtTracing** is both a control and status bit. Reading it indicates Event Trace is tracing or not. Writing to it enables native software to start or stop Event Trace dynamically such as turning it off at the beginning of a function and turning it back on at the end, or visa versa.
- Event Trace can be started by native software writing to the Event Trace control registers. This can also be done with a JTAG command writing to the same registers.
- Event Trace can be first enabled via (for example) a hardware trigger set up to match on a program instruction address then, when the program-to-be-traced starts executing and the trigger match occurs, it in turn sets **trTeEvtControl.trTeEvtTracing**, and event trace starts.
- Once running, a hardware trigger previously set up to disable trace can, at the occurrence of a trigger match, turn Event Trace off which clears **trTeEvtControl.trTeEvtTracing**. Similar actions are common with Instruction Trace.
- An instrumented program can also clear or set **trTeEvtControl.trTeEvtTracing** to disable or enable Event Trace.
- Event Trace can be disabled by writing **trTeEvtControl.trTeEnable** = 0. This does a flush to the trace system.
- Trace can be set up to stop tracing on buffer full: **trRamControl.trRamStopOnWrap** = 1. When this occurs, Event Trace continues running, however no more trace messages are written to the trace buffer.
- A self-hosted trace analysis program such as Linux *perf* could control the starting and stopping of Event Trace.
- Trace hardware supports on-the-fly switching between Event Trace and Instruction Trace. Memory-mapped (MMIO) hardware control registers allow a user program to change modes of trace so a trace recording can have both instruction trace messages intermixed with event trace messages. This feature allows the tracing of a portion of one program/process full execution while also tracing kernel-level events such as going in and out of kernel mode and running other processes.
- Extension to the Instruction Trace message (for N-trace this is ProgTraceCorrelation message) will generate a disable trace message for two additional conditions:
  - Inhibiting trace for security reasons
  - Unbounded wait - pausing trace for an unbounded amount of time (e.g. executing WFI or WFE instructions)

Appendix B: Event Trace Global and per-Event Filtering (non-normative)

These global and per-event filters are documented in the **RISC-V Trace Control Interface Specification** [3 pp. Ch. 6.3]. These are documented here since they are already documented and the specification ratified.

There are several types of useful Event Trace filters that can be provided by hardware to reduce the volume of trace data and focus on relevant information. While these registers and associated hardware are optional, they are required to be included in the implementation of Trace hardware.

Global filters are:

- Privilege level filtering - only trace events that occur in a specific privilege level or set of privilege levels. This is useful for tracing events in user mode vs. kernel mode.
- Context ID filtering - only trace events that occur in a specific context ID or set of context IDs. This is useful for tracing events in a specific process or thread.

Per-event filters are:

- Cause register filtering - only trace events that occur with specific cause register values are traced. These are applicable to exception or interrupt Event Types.

Below, the 'i' in trTeFilteriControl (and others) is reference to the index of the register (0, 1, etc), which means the developer can instantiate multiple sets of filter registers.

B.1. Privilege Level Filtering Requirements

Table 35. Privilege Level Filtering Registers

Register	Register Fields	Description
trTeFilteriControl	**.trTeFilterEnable** **.trTeFilterMatchPrivilege**	Set = 1 Set = 1 All other fields set to 0
trTeFilteriMatchInst	**.trTeFilterMatchChoicePrivilege**	[7:0] bits to match Privilege levels list in table below

Table 36. Privilege level encoding (from RHTI spec)

Value	Description
0	U
1	S/HS
2	reserved
3	M
4	D (debug mode)
5	VU
6	VS
7	reserved

B.2. Context ID Filtering Requirements

To enable Event Trace on matching a single Context ID write value and disable Event Trace on a non-match write to Context ID requires the Table setup below.

For **trTeCompjControl** and **trTeCompjMatchLow** below, j=0 which defines the first comparator. If the developer wants to support multiple Context ID matches, then j=1,2,... defines additional comparators.

Table 37. Context ID Filtering

Register	Register Fields	Description
trTeComp0Control	**.trTeCompPInput** **.trTeCompSInput**	[1:0] Set = 1: context [3:2] Set = 1: context
	.trTeCompPFunction **.trTeCompSFunction**	[6:4] Set = 0: **equal to** trTeCompPMatch [10:8] Set = 0: **equal to** trTeCompSMatch
	.trTeCompMatchMode	[13:12] Set = 0: **primary result true**
	trTeCompPNotify	Set = 1 to generate an Event Trace packet when primary comparator match occurs
	All other fields set to 0	
trTeComp0MatchLow	**.trTeCompPMatchLow**	[31:0] **Match value for** (primary comparator)

(Secondary comparator is not needed for context ID filtering since it is only a single value match. It is unclear whether these 'S' fields need to be included in the hardware).

B.3. Exception Cause Register Filtering

Table 38. Exception Cause (Ecause) Filtering

Register	Register Fields	Description
trTeFilteriControl	**.trTeFilterEnable** **.trTeFilterMatchEcause**	Set = 1 Set = 1 (to compare Ecause value) All other fields set to 0
trTeFilteriMatchInst	**.trTeFilterValueInterrupt** (Note the name doesn't match the comparison)	[8:8] Set = 0 to match on exception (itype = 1) All other fields set to 0
trTeFilteriMatchEcauseLow	**.trTeFilterMatchChoiceEcauseLow**	[31:0] bit mask; set bit(s) to 1 to match encoded value of Ecause
trTeFilteriMatchEcauseHigh	**.trTeFilterMatchChoiceEcauseHigh**	[63:32] bits of Ecause (high) value

Table of (encoded) Exception cause values (from RISC-V Instruction Set Manual: Vol II).

Table 39. Exception cause values

Value	m-cause	s-cause	vs-mode	Value	m-cause	s-cause	vs-mode
0	Instruction address misaligned	Instruction address misaligned	Instruction address misaligned	13	Load page fault	Load page fault	Load page fault
1	Instruction access fault	Instruction access fault	Instruction access fault	14	Reserved	Reserved	Reserved
2	Illegal instruction	Illegal instruction	Illegal instruction	15	Store/AMO page fault	Store/AMO page fault	Store/AMO page fault
3	Breakpoint	Breakpoint	Breakpoint	16	Double trap	Reserved	Reserved
4	Load address misaligned	Load address misaligned	Load address misaligned	17	Reserved	Reserved	Reserved
5	Load access fault	Load access fault	Load access fault	18	Software check	Software check	Reserved
6	Store/AMO address misaligned	Store/AMO address misaligned	Store/AMO address misaligned	19	Hardware error	Hardware error	Reserved
7	Store/AMO access fault	Store/AMO access fault	Store/AMO access fault	20	Reserved	Reserved	Instruction guest-page fault
8	Environment call from U-mode	Environment call from U-mode	Environment call from U-mode or VU-mode	21	Reserved	Reserved	Load guest-page fault
9	Environment call from S-mode	Environment call from S-mode	Environment call from HS-mode	22	Reserved	Reserved	**Virtual instruction**
10	Reserved	Reserved	Environment call from VS-mode	23	Reserved	Reserved	Store/AMO guest-page fault
11	Environment call from M-mode	Reserved	Environment call from M-mode	24-31	Custom use	Custom use	Custom use
12	Instruction page fault	Instruction page fault	Instruction page fault	32-47	Reserved	Reserved	Reserved
				48-63	Custom use	Custom use	Custom use

B.4. Interrupt Cause Register Filtering

Several of the registers are "shared" with Ecause filtering, thus the name of the field may not match the comparison being made. For example, trTeFilterMatchChoiceEcauseLow is used to compare against the cause value for both exceptions and interrupts.

Table 40. Interrupt Cause Register Filtering

Register	Register Fields	Description
trTeFilteriControl	**.trTeFilterEnable** **.trTeFilterMatchInterrupt**	Set = 1 Set = 1 (to compare Interrupt cause value) All other fields set to 0
trTeFilteriMatchInst	**.trTeFilterValueInterrupt**	[8:8] Set = 1 to match on interrupt (itype = 2) All other fields set to 0
trTeFilteriMatchEcauseLow	**.trTeFilterMatchChoiceEcauseLow**	[31:0] bit mask; set bit(s) to 1 to match encoded values 0 to 31 of Interrupt cause
trTeFilteriMatchEcauseHigh	**.trTeFilterMatchChoiceEcauseHigh**	[63:32] bit mask; set bit(s) to 1 to match encoded values 32 to 63 of Interrupt cause

The (encoded) Interrupt cause values (from the RISC-V Instruction Set Manual: Vol II) are located in Table 16 - mcause, Table 37 - scause, and Table 50 when the hypervisor extension is implemented. Below is Table 50:

Exception Code	Description
0	Reserved
1	Supervisor software interrupt
2	Virtual supervisor software interrupt
3	Machine software interrupt
4	Reserved
5	Supervisor timer interrupt
6	Virtual supervisor timer interrupt
7	Machine timer interrupt
8	Reserved
9	Supervisor external interrupt
10	Virtual supervisor external interrupt
11	Machine external interrupt
12	Supervisor guest external interrupt
13	Counter-overflow interrupt
14-15	Reserved
≥16	Designated for platform use

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